

Inferred Transformations for Structure-Compatible Generalization

Jawaun Brown

Abstract

Phase one showed that oracle transformation compatibility predicts OOD behavior under underspecification. This phase asks whether the oracle can be weakened: infer supported transformations from training evidence, score compatibility under that discovered family, and use the discovered family as a train-time regularizer. In a modular addition domain, a shift is admitted only when observed train-label overlaps support the induced input and label action. The resulting suite evaluates both prediction and control without using OOD labels for model selection or training.

1. Regime Transition

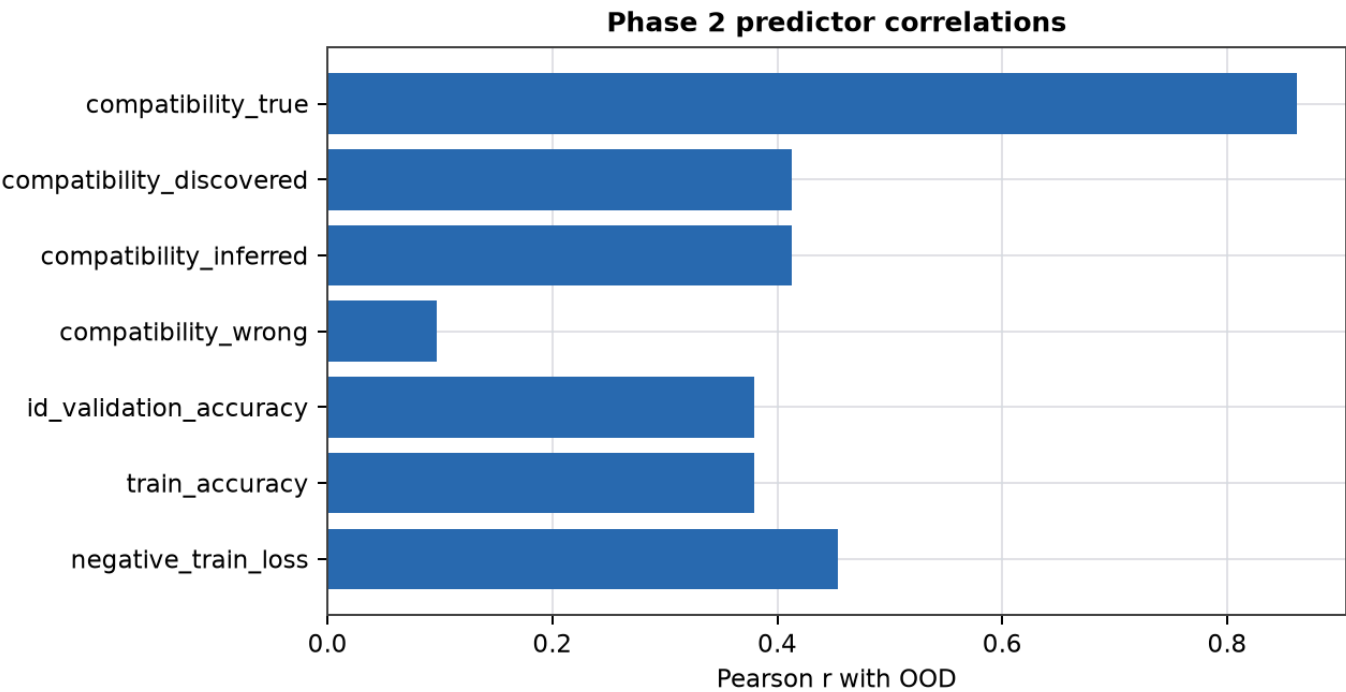
The old regime supplied the deployment group. The new regime infers a supported finite transformation family, rejects vacuous shifts, and applies compatibility pressure to unlabeled domain points.

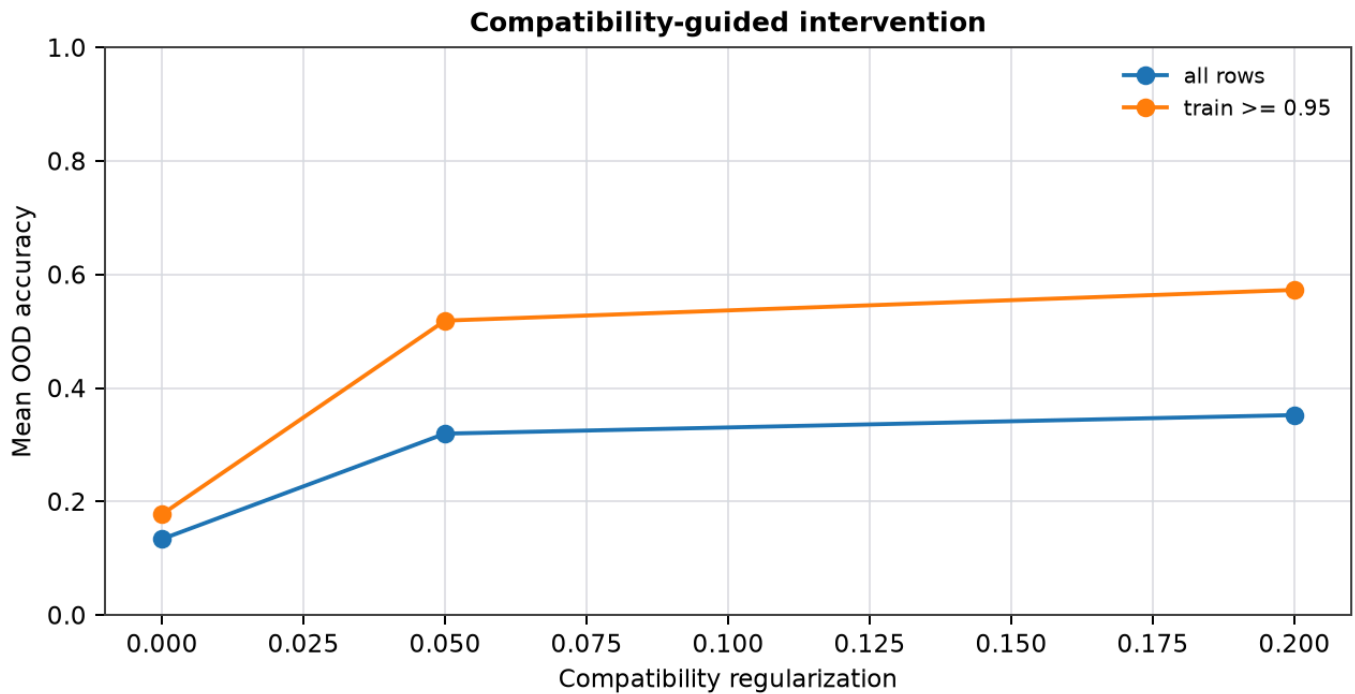
2. Result

The strongest phase-two predictor was `compatibility_true` (Pearson $r=0.862$ with OOD).

The best high-ID regularization arm was 0.200, with high-ID mean OOD 0.573. Delta versus zero regularization: 0.395.

Figures





3. Scope

This is a finite modular-domain intervention result. It supports the broader OOD-certifiability-lite program, but it does not yet solve transformation discovery for vision, language, or open deployment shifts.

4. Architecture Lesson

The architecture change is small: infer which transformations are supported by the training evidence, reject vacuous shifts, and use the accepted family as a compatibility regularizer on unlabeled domain points. In virtual-governor language, this converts a system-level deployment constraint into local training pressure. The result is not that the model understands the governor; it is that the training loop exposes and controls one governing stress.

5. Next Operation

The next operation is to make the discovery family learned rather than enumerated, then transfer the same intervention protocol to vision rotations and paraphrase/template substitutions.